ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 36 g, 1 : 8	4 + 32 = 36; 4:32 = 1:8.
3	**(c)** homogeneous mixture	Alloy.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(d)** 1 g and 8 g	Ratio 1 : 8.
6	**(d)** hydrogen gas	Metal with acid.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** 14	2 x 7.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R is the definition of an element.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but malleability and ductility do not explain conduction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oxygen = 36 - 4 = 32 g (1); Ratio = 4 : 32 = 1 : 8 (1).	1+1
12	They have a lower melting point than the metals (1); Allow easy joining by melting (1).	1+1
13	Their outer arrangement is stable (1); Used where reaction is unwanted, as in bulbs (1).	1+1
14	One molecule has 3 atoms (1); 3 x 3 = 9 atoms (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Probably iron, an element (1); Magnetic and reacts with acid giving hydrogen (1); Could also be a mixture containing iron (1).	1+1+1
16	In diamond all electrons are held in fixed bonds (1); In graphite some electrons are free to move (1); Different arrangements (1).	1+1+1
17	A homogeneous mixture (1); Evaporate and condense water (1); Distillation (1).	1+1+1
	OR: Sodium: soft reactive metal (1); Chlorine: poisonous greenish gas (1); NaCl: harmless white crystalline solid (1).	1+1+1
18	All dissolve completely (1); Uniform composition (1); Components cannot be seen separately (1).	1+1+1
	OR: Heat sugar in a dry tube (1); Black residue forms (1); Carbon is present (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Iron filings (1). (ii) Sand (residue) (1). (iii) Common salt (1). (iv) Magnetic separation, filtration and evaporation (1); the substances keep their own properties and are not chemically combined, and they were separated by physical methods (1).	1+1+1+2
20	Compound has a fixed mass ratio (1); Hydrogen = 45 / 9 = 5 g (1); Oxygen = 40 g (1); Same ratio whatever the source (1); Mixtures do not obey this (1).	1+1+1+1+1
	OR: Metal oxides are basic (1); Non-metal oxides are acidic (1); Metals react with acids to give hydrogen (1); Physical: lustre, ductility, sonority (1); Exceptions need care (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a molecule such as O_2 a compound; a compound needs *different* elements.
- Saying air or an alloy is a compound; they are mixtures.

- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.
- Writing that all non-metals are poor conductors; graphite is an exception.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $3O_2$	Coefficient before formula.
2	**(c)** sulphur	SO2.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R is a supporting reason.
4	**(c)** a compound copper oxide formed	Copper reacts with oxygen.
5	**(a)** shows properties of both metals and non-metals	Intermediate.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(c)** a metal	Metallic properties.
8	**(d)** semiconductor	Controlled conduction.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Kerosene keeps air and moisture away from sodium.
10	**(c)** A is true but R is false.	A is true (iron is attracted by a magnet); R is false: compounds need chemical methods.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It does not rust (1); It is strong and easy to clean (1).	1+1
12	Distilled water is pure; same throughout (1); Milk is a mixture of several substances (1).	1+1
13	In a mixture each substance keeps its properties (1); The compound is a new black substance (1).	1+1
14	It is less dense and reactive (1); Produces hydrogen and heat (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	H2O: 2 H and 1 O (1); H2O2: 2 H and 2 O (1); Different formulae, different compounds (1).	1+1+1
16	Dazzling white flame (1); White powder forms (1); Magnesium oxide, a compound (1).	1+1+1
17	Highly reactive (1); Combine with oxygen and water (1); Found as compounds (1).	1+1+1
	OR: Copper = 450 g (1); Tin = 50 g (1); Total 500 g (1).	1+1+1
18	Rusting is chemical and forms iron oxide (1); Dissolving is physical (1); Sugar can be recovered (1).	1+1+1
	OR: Mg gives basic oxide (1); S gives acidic oxide (1); Tested with litmus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) P (1). (ii) Sonority (1). (iii) Graphite (carbon) (1). (iv) Brittle, poor conductor of heat, not lustrous, not malleable/ductile (any two) (2).	1+1+1+2
20	Magnet to remove iron (1); Filter to remove sand (1); Evaporate to obtain salt (1); Condense vapour for water (1); Physical properties used: magnetism, particle size, boiling point (1).	1+1+1+1+1
	OR: Elements are also pure substances (1); Compounds are pure with fixed composition (1); Examples: iron, oxygen (elements) (1); Water, salt (compounds) (1); Mixtures are not pure (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a molecule such as O_2 a compound; a compound needs *different* elements.
- Saying air or an alloy is a compound; they are mixtures.
- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.

- Writing that all non-metals are poor conductors; graphite is an exception.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** constituents cannot be separated by simple physical methods	Chemical bonds.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(d)** blue litmus red	Acidic.
4	**(d)** a suspension (heterogeneous)	Particles settle.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** wood	Non-metal.
7	**(d)** electrolysis of water	Chemical decomposition.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(d)** A is false but R is true.	Water is a compound, so A is false; R is true.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Sulphur is a typical non-metal showing these properties.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Non-metal (1); e.g. sulphur gives sulphur dioxide (1).	1+1
12	C: 6, H: 12, O: 6 (1); Total 24 atoms (1).	1+1
13	Metalloids show mixed properties (1); Exceptions like graphite and mercury exist (1).	1+1
14	Iron displaces copper (1); Chemical change (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Brass is an alloy of copper and zinc, unreactive with moisture (1); Iron combines with oxygen and water to rust (1); Alloying changes properties (1).	1+1+1
16	They contain more than one metal (1); Composition varies (1); They are alloys, i.e. mixtures (1).	1+1+1
17	Copper conducts (1); Plastic insulates (1); Prevents shocks (1).	1+1+1
	OR: Magnet removes iron: magnetic property (1); Heat the rest: ammonium chloride sublimes and is collected (1); Sand remains behind (1).	1+1+1
18	Mercury is a liquid metal (1); Bromine is a liquid non-metal (1); Gallium melts in the hand (1).	1+1+1
	OR: Contain many components (1); Composition varies (1); Components keep properties (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mixture (1). (ii) Iron sulphide (1). (iii) No (1). (iv) In the mixture the components keep their properties and can be separated physically; the compound has new properties and cannot be separated by a magnet; heat/energy change occurs on forming the compound (any two) (2).	1+1+1+2
20	Dissolving: physical, mixture (1); Heating iron-sulphur: chemical, compound (1); Rusting: chemical (1); Melting wax: physical (1); Physical changes are reversible, chemical form new substances (1).	1+1+1+1+1
	OR: Heat the mixture (1); Ammonium chloride sublimes (1); Collect the vapour on a cold surface (1); Sand remains (1); Property: sublimation (1).	1+1+1+1+1

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- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.
- Writing that all non-metals are poor conductors; graphite is an exception.

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